

Practical No. 2

Name : Danish Altaf Solkar

Class : CO - B

Enrolment no. : 24111590805

Batch : 2

Aim : Select relevant process model to define activities and related tasks set

1. Aim

To select a suitable software process model for a robotics and drone-based emergency response system and identify the major software development activities and related tasks involved in the project.

2. Project Name

Smart Emergency Response Drone and Robot System

3. Project Description

The Smart Emergency Response Drone and Robot System is a software-based solution designed to support pressing challenges such as medical emergencies and search and rescue operations. The system can coordinate drones and robots to collect information from difficult or unsafe areas, assist in locating people, and support emergency teams with timely information.

4. Project Objectives

- To support medical emergency response operations.
- To assist in search and rescue operations.
- To remotely monitor difficult or unsafe areas.
- To collect and transmit information from drones and robots.
- To help emergency teams identify and respond to situations quickly.
- To provide a software interface for monitoring and controlling robotic devices.
- To improve coordination between field devices and emergency personnel.

5. Target Users

- Emergency Response Teams
- Medical/Emergency Personnel
- Search and Rescue Teams
- System Administrator

6. Major Features

Module	Major Features
Emergency Team	Login, monitor devices, receive alerts, view live/status information, coordinate response
Drone Module	Device control, location tracking, image/video or sensor data collection, emergency-area monitoring
Robot Module	Remote control, navigation/status monitoring, sensor data collection, assistance in unsafe areas
Emergency Management	Create emergency operation, assign devices, track mission status, manage alerts
Administrator	Manage users, devices, system settings and reports

7. Project Requirements

Functional Requirements

- User authentication and authorization.
- Drone and robot registration and management.
- Remote monitoring and control of devices.
- Emergency alert and notification management.
- Location and mission tracking.
- Collection and display of device/sensor information.
- Mission status and report generation.

Non-Functional Requirements

- Security
- Reliability
- Performance
- Usability
- Availability
- Scalability

8. Selected Software Process Model

Scrum – Agile Process Model

Scrum is selected because a robotics and drone emergency-response system may require changing requirements, frequent testing, hardware/software integration, and continuous feedback from emergency-response users. The system can be developed incrementally through short sprints.

9. Reasons for Selecting Scrum

- Requirements can change during development.
- The project can be divided into smaller increments.
- Working features can be delivered after each sprint.
- Emergency-response users can provide continuous feedback.
- Software and device integration can be tested continuously.
- Problems can be identified and corrected early.
- New features and improvements can be prioritized through the product backlog.

10. Scrum Activities and Related Tasks

Activity	Related Tasks	Deliverable / Output
Product Backlog Creation	Identify stakeholders, collect requirements, list features, prioritize requirements	Product Backlog
Sprint Planning	Select features, estimate tasks, define sprint goal	Sprint Backlog
Design	Design user interface, database, communication and system architecture	Design Document
Development	Develop control/monitoring modules, implement features, integrate components	Working Features
Testing	Unit, integration, system and device-integration testing	Test Results / Report
Sprint Review	Demonstrate completed features and collect user feedback	Review Feedback
Sprint Retrospective	Identify problems and improvements	Improvement Actions
Deployment	Configure system and release working increment	Working System
Maintenance	Fix bugs and implement approved enhancements	Updated System

11. Example Sprint Plan

Sprint	Major Features
Sprint 1	User registration, login and emergency-team management
Sprint 2	Drone and robot registration and status monitoring
Sprint 3	Device control, location tracking and communication
Sprint 4	Emergency mission management and alerts
Sprint 5	Search and rescue monitoring and data collection
Sprint 6	Reports, security, testing and improvements

12. Scrum Process Diagram

PRODUCT BACKLOG

↓

SPRINT PLANNING

↓

SPRINT: Development + Daily Scrum + Testing

↓

SPRINT REVIEW

↓

SPRINT RETROSPECTIVE

↓

WORKING PRODUCT INCREMENT

↓

UPDATED PRODUCT BACKLOG → NEXT SPRINT

13. Activity → Task → Deliverable

Activity	Task	Deliverable
Requirement Analysis	Collect and validate emergency-response requirements	SRS
Planning	Estimate time, cost and resources	Project Plan
Design	Prepare UI, database, communication and system architecture	Design Document
Development	Coding, device integration and implementation	Source Code / Working Module
Testing	Unit, integration, system and device-integration testing	Test Report
Deployment	Install and configure the application/system	Working System
Maintenance	Bug fixing and enhancement	Updated System

14. Conclusion

For the Smart Emergency Response Drone and Robot System, the Scrum process model was selected because the project may have changing requirements, requires continuous testing and integration, and benefits from regular feedback from emergency-response users. The project can be divided into multiple sprints, with each sprint delivering a working set of features. The identified activities include backlog creation, sprint planning, design, development, testing, review, retrospective, deployment and maintenance.